

1) H_{cf} is always real valued :

$$\left. \begin{array}{l} |X_{cf}|^2 \in \mathbb{R} \\ |D_{cf}|^2 \in \mathbb{R} \end{array} \right\} H_{cf} \in \mathbb{R}$$

$$2) \quad D_{CFI} \rightarrow 0 \quad \Rightarrow \quad H_{CFI} = \frac{|X_{CFI}|^2}{|X_{CFI}|^2} = 1$$

$$D_{CFI} \rightarrow \infty \quad \Rightarrow \quad H_{CFI} = \frac{|X_{CFI}|^2}{|X_{CFI}|^2 + \infty} = 0$$

$$H_{CFI} \leq 1$$

$$\frac{|X_{CFI}|^2}{|X_{CFI}|^2 + |D_{CFI}|^2} \leq 1$$

$$| \cdot (|X_{CFI}|^2 + |D_{CFI}|^2)$$

$$|X_{CFI}|^2 \leq |X_{CFI}|^2 + |D_{CFI}|^2 \quad | - |X_{CFI}|^2$$

$$0 \leq |D_{CFI}|^2 \quad \text{always true}$$

$$H_{CFI} \geq 0$$

$$\frac{|X_{CFI}|^2}{|X_{CFI}|^2 + |D_{CFI}|^2} \geq 0$$

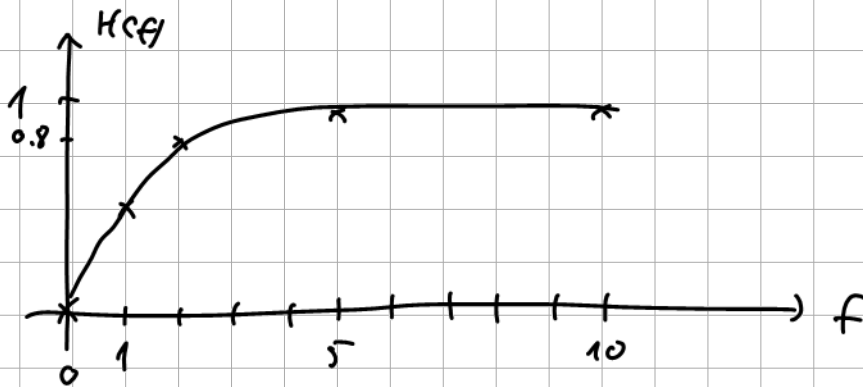
$$| \cdot (|X_{CFI}|^2 + |D_{CFI}|^2)$$

$$|X_{CFI}|^2 \geq 0 \quad \text{always true}$$

$$3) \quad X(f) = f$$

$$|X(f)|^2 = f^2$$

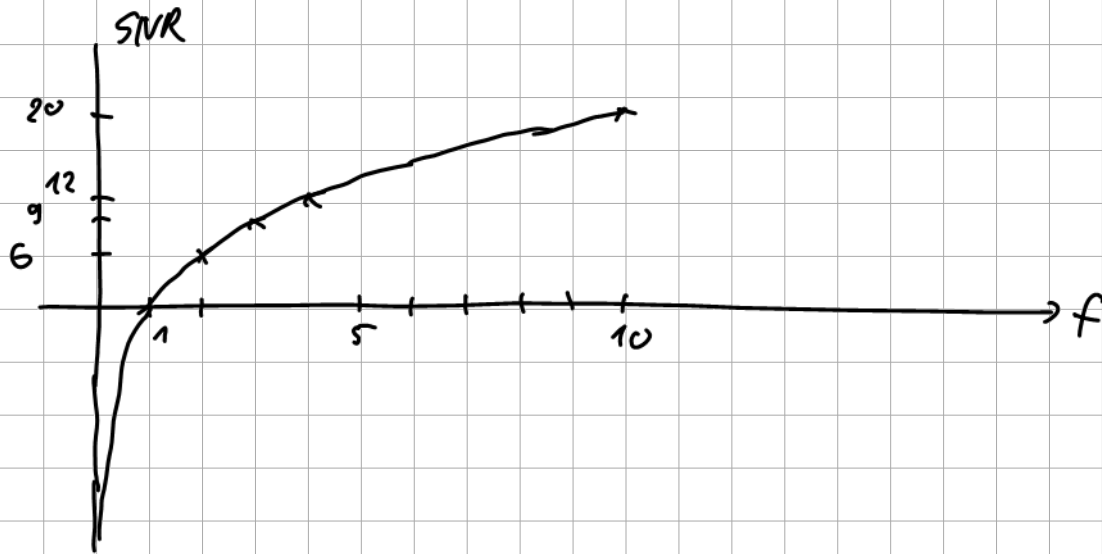
$$H(f) = \frac{f^2}{f^2 + 1}$$



4)

$$SNR = 10 \log_{10} \frac{|X_{cf}|^2}{|D_{cf}|^2}$$

$$= 10 \log_{10} f^2$$



5) The Denominator is greater or equal zero.

The addition of a small positive number prevents a division by zero without affecting non-zero elements of the denominator too much.

$$6) \text{ Hoelder mean} = \left(\frac{1}{N} \sum_n |x_n|^p \right)^{\frac{1}{p}}, \text{ here } p = -1$$

$$x_n = (1 \quad 2 \quad 3)$$

$$\Rightarrow \left(\frac{1}{3} \left(|1|^{-1} + |2|^{-1} + |3|^{-1} \right) \right)^{\frac{1}{-1}}$$

$$= \left(\frac{1}{3} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} \right) \right)^{-1} = \frac{3}{\underbrace{1 + \frac{1}{2} + \frac{1}{3}}_{\text{harmonic mean}}} = \frac{3}{\frac{6}{6} + \frac{3}{6} + \frac{2}{6}} = \frac{18}{11}$$







